Database Handling

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- **SQLite** is an **open-source relational database** i.e. used to perform database operations on android devices such as storing, manipulating or retrieving persistent data from the database.
- It is **embedded** in android by default.
- So, there is no need to perform any database setup or administration task.

SQLiteDatabase class

It contains methods to be performed on sqlite database such as create, update, delete, select etc.

Method	Description
void execSQL(String sql)	executes the sql query not select query.
long insert(String table, String nullColumnHack, ContentValues values)	inserts a record on the database. The table specifies the table name, nullColumnHack doesn't allow completely null values. If second argument is null, android will store null values if values are empty. The third argument specifies the values to be stored.
int update(String table, ContentValues values, String whereClause, String[] whereArgs)	updates a row.
Cursor query(String table, String[] columns, String selection, String[] selectionArgs, String groupBy, String having, String orderBy)	returns a cursor over the resultset.

Database Creation

- In order to create a database you just need to call this method **openOrCreateDatabase** with your database name and mode as a parameter.
- It returns an instance of SQLite database which you have to receive in your own object.
- Its syntax is given below

```
SQLiteDatabase mydatabase = openOrCreateDatabase("your
database name",MODE_PRIVATE,null);
```

Database Fetching

- We can retrieve anything from database using an object of the **Cursor class**.
- We will call a method of this class called **rawQuery** and it will return a **resultset** with the cursor pointing to the table.
- We can move the cursor forward and retrieve the data.

Database Fetching

Sr.No	Method & Description
1	**getColumnCount()** This method return the total number of columns of the table.
2	**getColumnIndex(String columnName)** This method returns the index number of a column by specifying the name of the column
3	**getColumnName(int columnIndex)** This method returns the name of the column by specifying the index of the column
4	**getColumnNames()** This method returns the array of all the column names of the table.
5	**getCount()** This method returns the total number of rows in the cursor
6	**getPosition()** This method returns the current position of the cursor in the table
7	**isClosed()** This method returns true if the cursor is closed and return false otherwise

Activity_main.xml

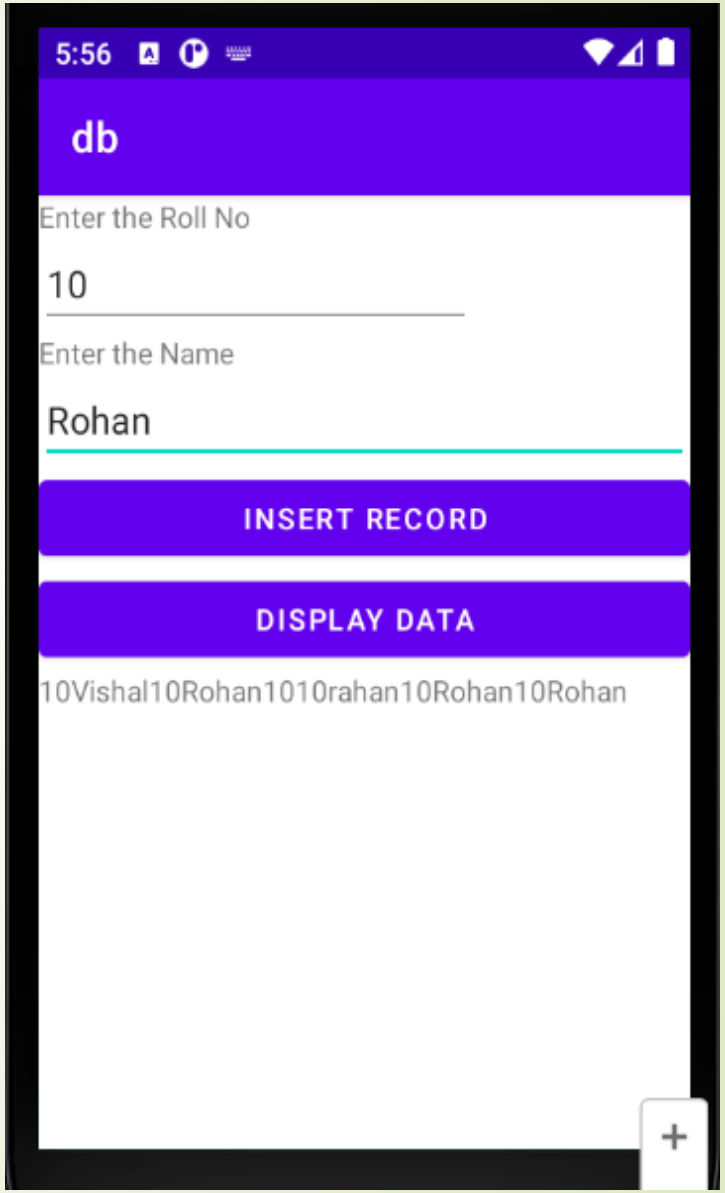
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Enter the Roll No" />
    <EditText
        android:id="@+id/editText1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPersonName"
        android:text="" />
```

```
<TextView
    android:id="@+id/textView2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Enter the Name" />
<EditText
    android:id="@+id/editText2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="textPersonName"
    android:text="" />
<Button
    android:id="@+id/button1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Insert Record" />
```

```
<Button
    android:id="@+id/button2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Display data" />
<TextView
    android:id="@+id/textView3"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="" />
</LinearLayout>
```

MainActivity.java

```
package com.example.db;
import android.content.ContentValues;
import android.database.Cursor;
import android.database.SQLException;
import android.database.sqlite.SQLiteDatabase;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    SQLiteDatabase db;
    Button b1, b2;
    TextView t1;
    String str="";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

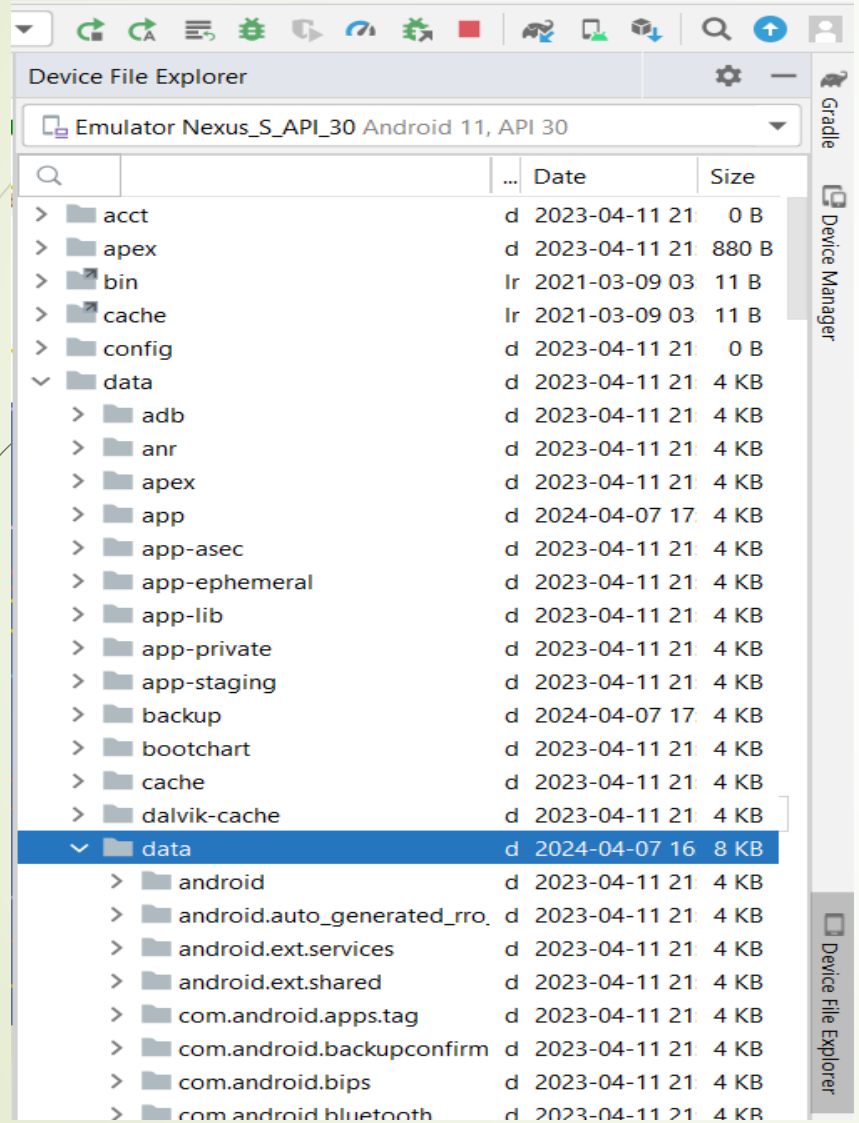


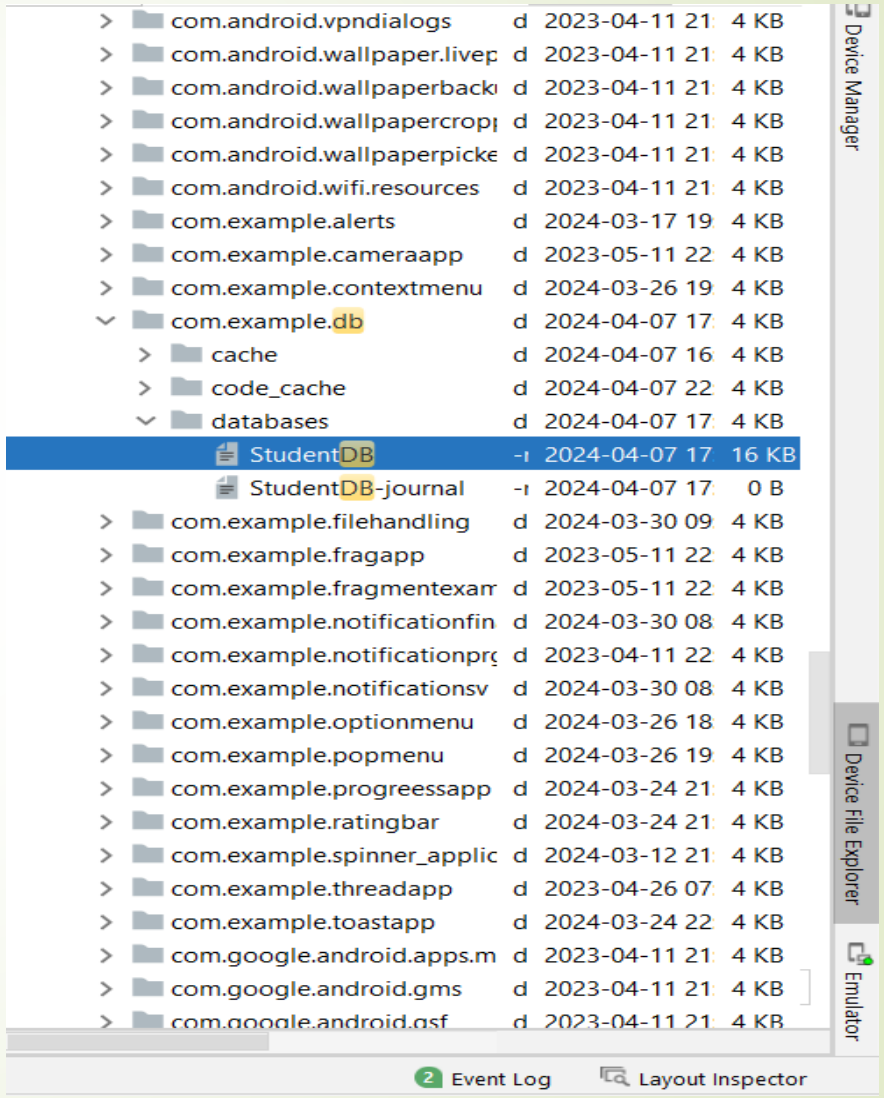
```
b1 = (Button) findViewById(R.id.button1);
b2 = (Button) findViewById(R.id.button2);
t1=(TextView) findViewById(R.id.textView3);
try
{
db = openOrCreateDatabase("StudentDB", SQLiteDatabase.CREATE_IF_NECESSARY,
                                                                    null);
db.execSQL("Create Table Temp(id integer,name text)");
 }
catch (SQLException e)
{
 }
```

```
b1.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
  // TODO Auto-generated method stub
   EditText eid = (EditText) findViewById(R.id.editText1);
   EditText ename = (EditText) findViewById(R.id.editText2);
   ContentValues values = new ContentValues();
   values.put("id", eid.getText().toString());
   values.put("name", ename.getText().toString());
   if ((db.insert("temp", null, values)) != -1) {
                Toast.makeText(MainActivity.this, "Record Successfully Inserted", 2000).show();
   }
    else {
                Toast.makeText(MainActivity.this, "Insert Error", 2000).show();
            }
    }});
```

```
b2.setOnClickListener(new View.OnClickListener() {
 @Override
   public void onClick(View v) {
            Cursor c = db.rawQuery("SELECT * FROM Temp", null);
            c.moveToFirst();
            while (!c.isAfterLast()) {
                str = str + c.getString(0) + c.getString(1);
                c.moveToNext();
            }
            t1.setText(str);
            c.close();
        }
     });
   }
 }
```

Database Location

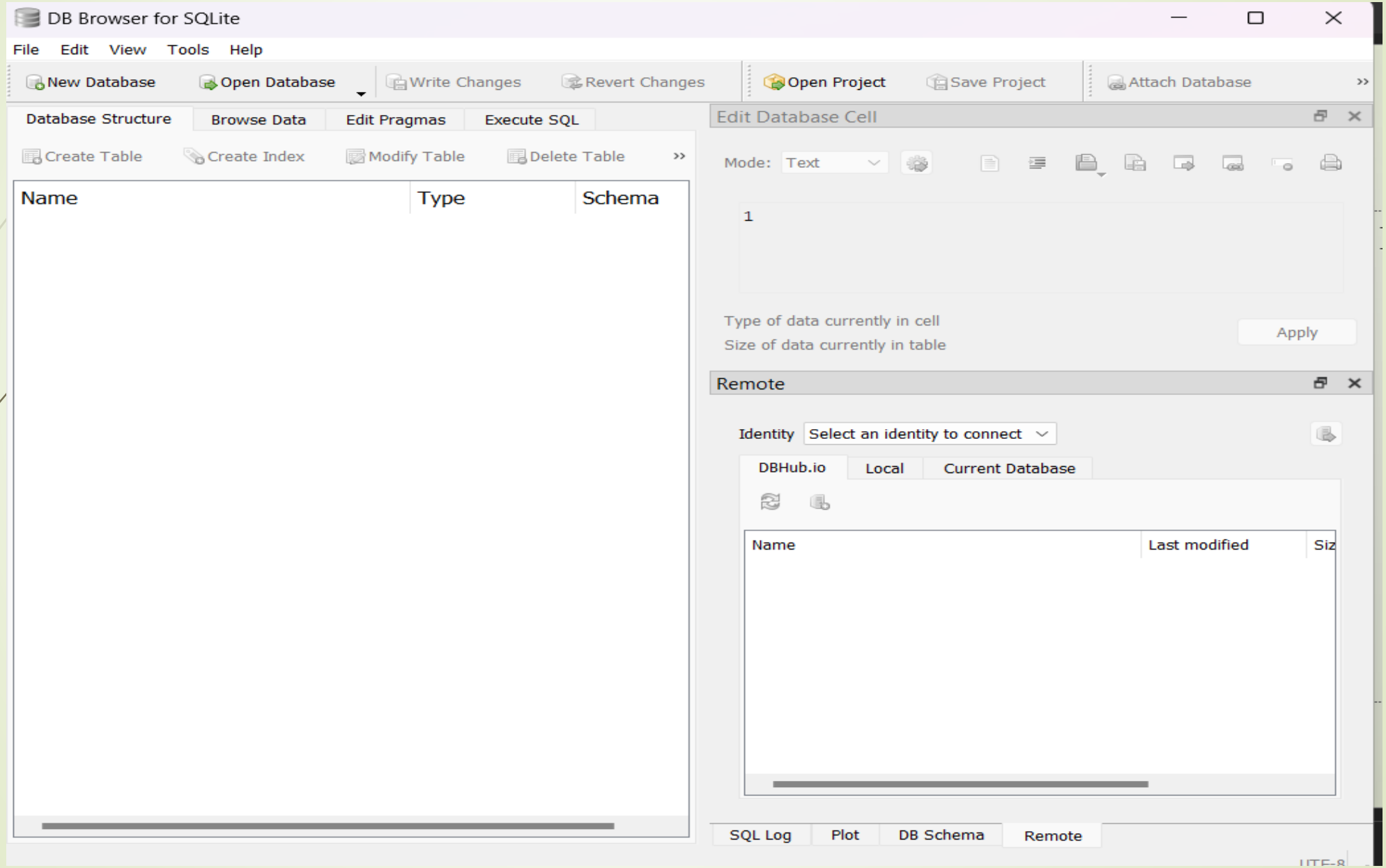




Database Opening

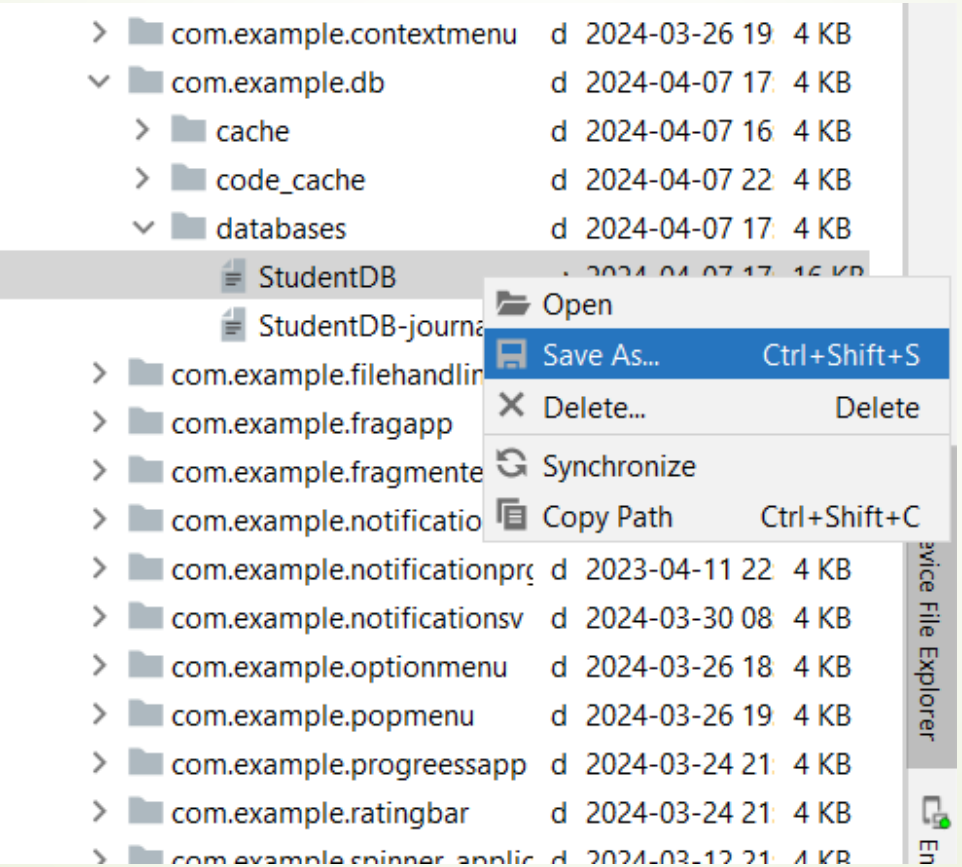
- Download any SqLite browser plugins or tool (in my case DB Browser for SQLite).
- **https://sqlitebrowser.org/dl/**
- DB Browser for SQLite - Standard installer for 64-bit Windows
- Go to Browse Data -> select your table (contacts) you will see the data stored.

Database Opening



Database Opening

C:\Users\ktmdy\AppData\Local\Google\AndroidStudio2021.1\device-explorer\Nexus_S_API_30 [emulator-5554]\data\data\com.example.db\databases\StudentDB



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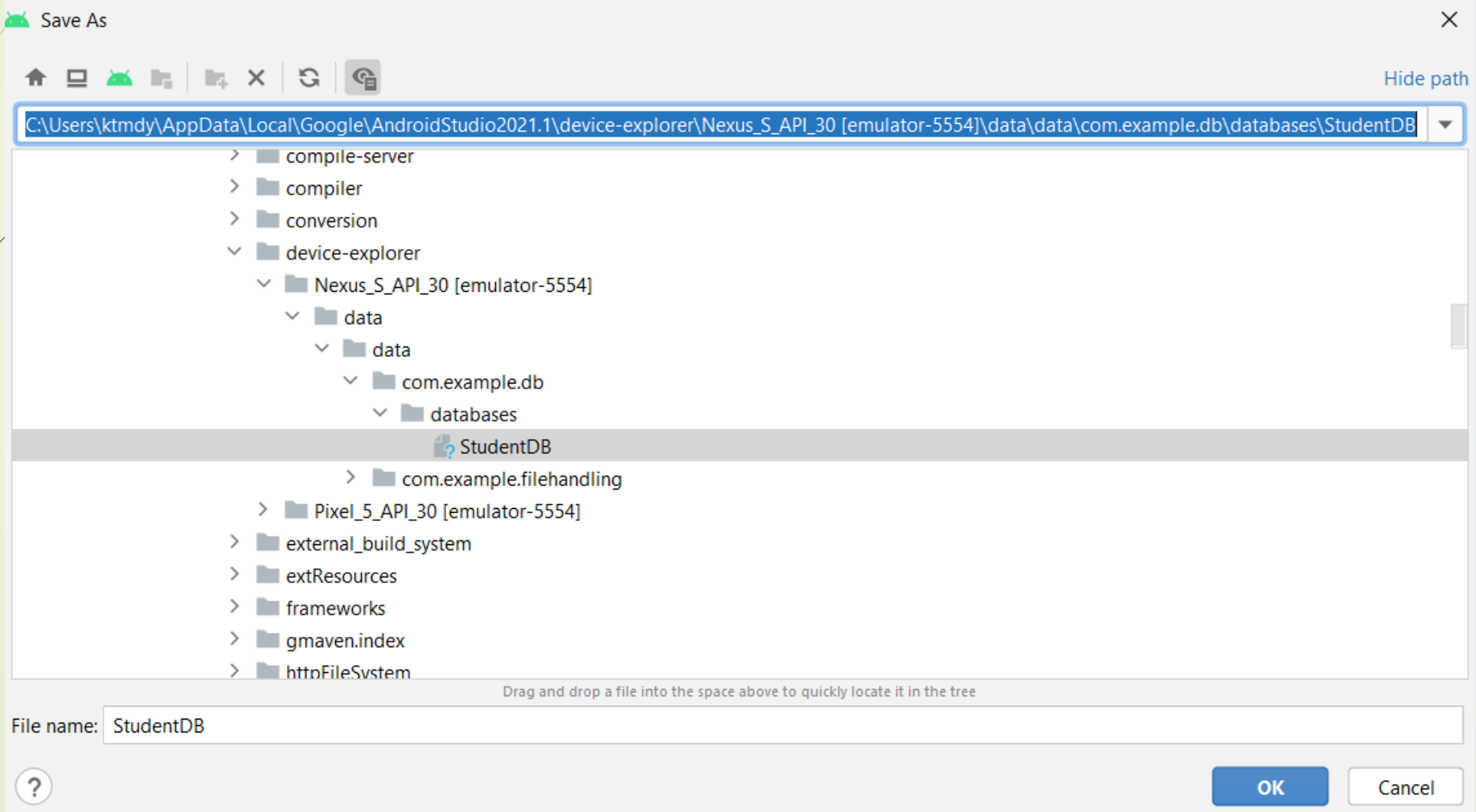

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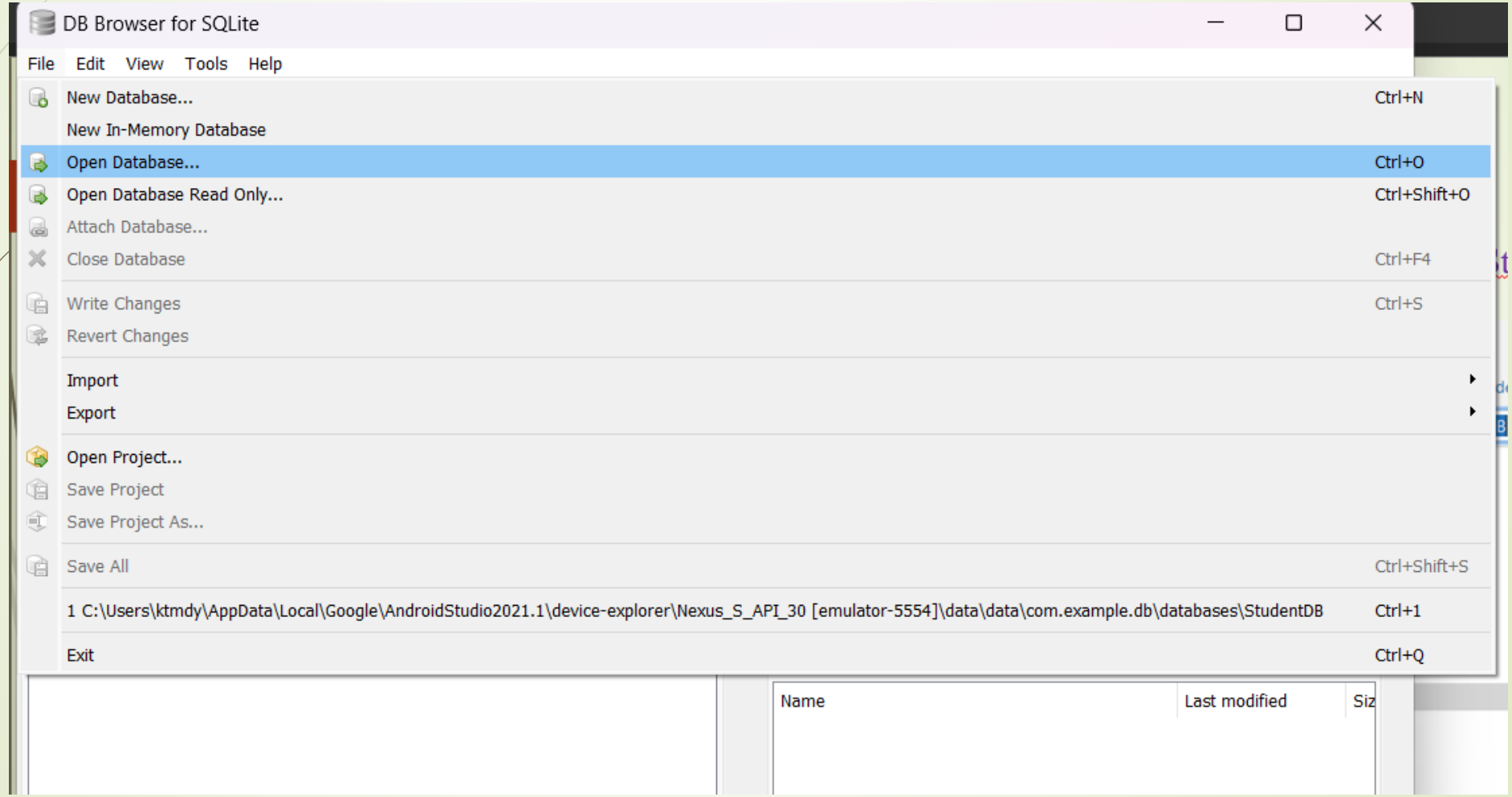

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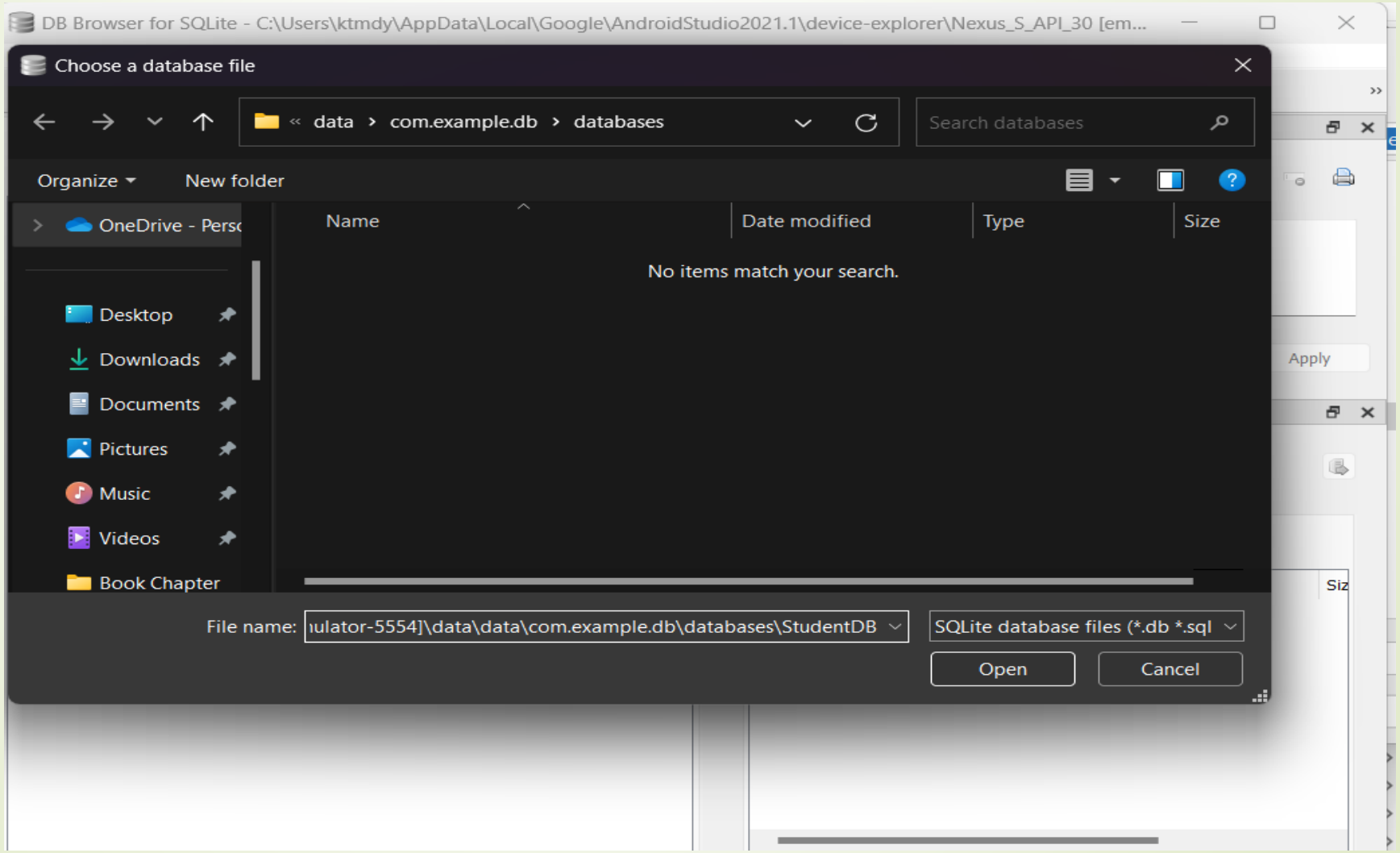

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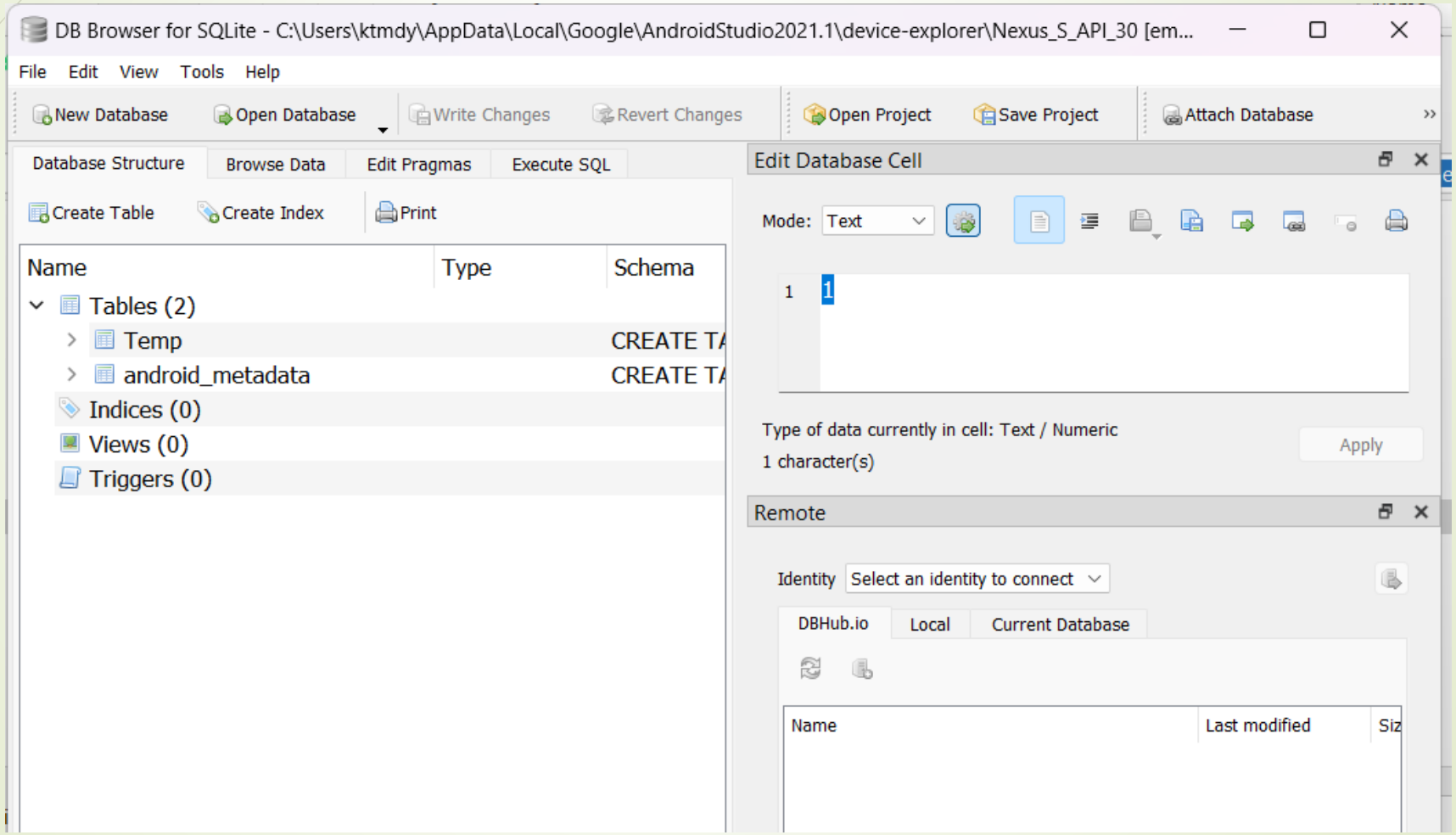

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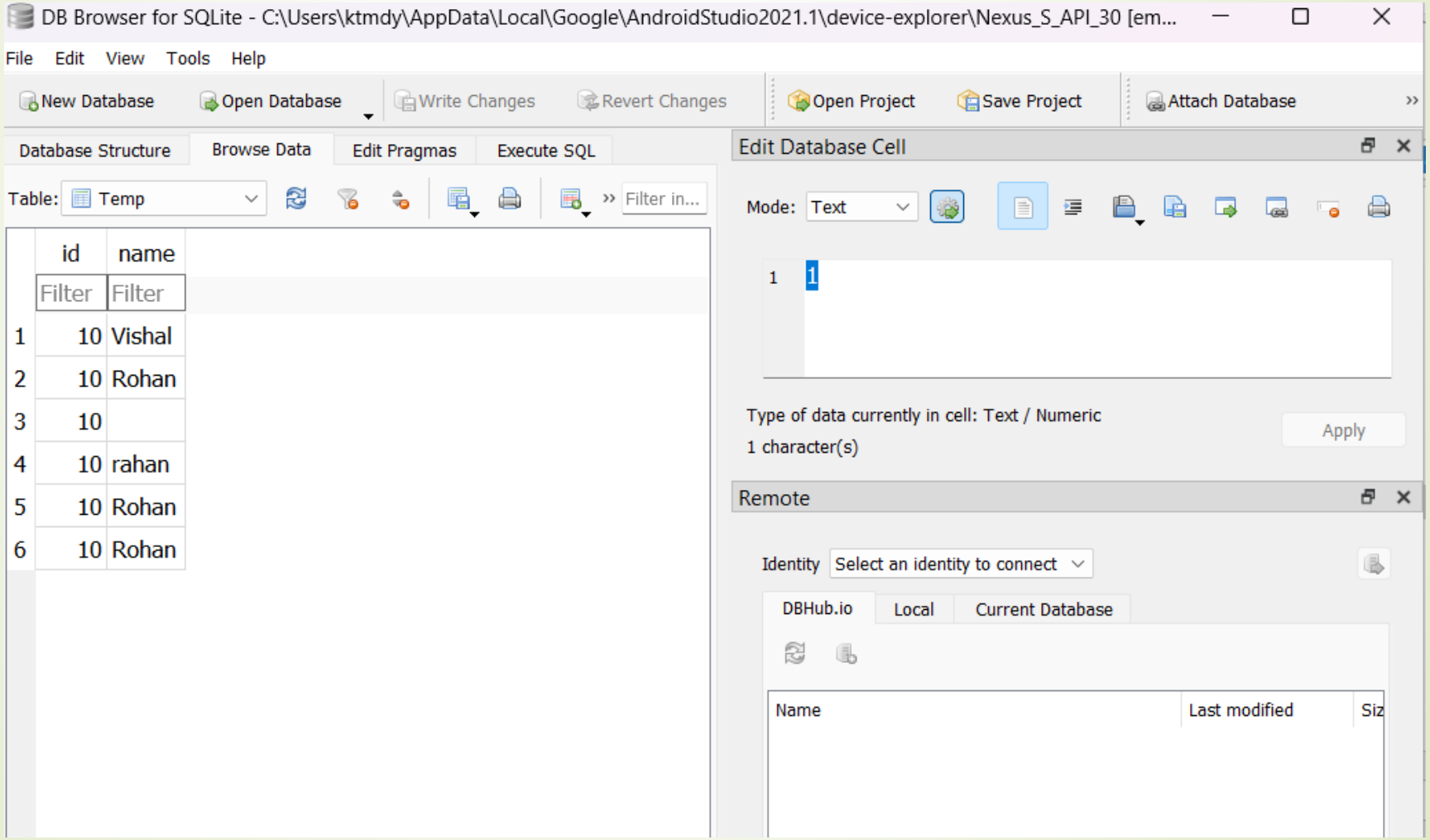
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